

Hemanth Kavula

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with over five years of experience building reliable data platforms across financial services, healthcare, and retail environments. Skilled in Python, SQL, PySpark, Databricks, SQL Server, SSIS, Snowflake, and cloud-based ETL/ELT development using AWS and Azure technologies. Experienced in **data architecture**, **dimensional modeling**, master and reference data integration, performance tuning, and developing governed data warehouses that provide trusted information for critical business decisions. Applies **data validation**, **unit testing**, **automated integration testing**, lineage, and quality frameworks to improve platform stability and reliability. At Fidelity Investments, collaborates with architects, analytics teams, and business leaders to deliver secure trading and market-data solutions, optimize database and Spark workloads, and establish reusable datasets that support portfolio analytics, risk monitoring, and regulatory reporting.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, Scala, PySpark

ETL/ELT & Orchestration: Microsoft SQL Server Integration Services (SSIS), Apache Airflow, dbt

Databases & Data Warehousing: SQL Server, Snowflake, PostgreSQL, MySQL, Stored Procedures

AWS Data Platform: Amazon S3, Amazon EMR, AWS Glue, Amazon Kinesis, Amazon Redshift, AWS Lambda, AWS

Azure Data Platform: Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Event Hubs, Microsoft Purview, Azure Synapse Analytics, Microsoft Azure

Big Data & Streaming: Apache Spark, Databricks, Delta Lake, Apache Kafka, Spark Structured Streaming

Data Modeling & Architecture: Dimensional Modeling, Star Schema, Slowly Changing Dimensions, Medallion Architecture, Master Data Management, Data Architecture

Data Quality & Testing: Great Expectations, Data Validation, Unit Testing, Automated Integration Testing, Data Lineage

Performance & DevOps: Query Optimization, Indexing, Spark Performance Tuning, Terraform, Jenkins, Git

Business Intelligence: Power BI, Tableau

PROFESSIONAL EXPERIENCE

Fidelity Investments | Jersey City, NJ

Jan 2025 – Present

Senior Data Engineer

- Designed and maintained **AWS Glue** and **Amazon EMR** ETL/ELT pipelines using PySpark to process trading and market data for investment risk, compliance, and regulatory reporting.
- Developed **SQL Server** data models, complex SQL queries, **stored procedures**, and SSIS workflows that integrated master and reference data into a trusted single source of truth.
- Built **Snowflake** dimensional models using **Star Schema** and SCD Type 2 patterns to support portfolio analytics, risk exposure analysis, and investment reporting.
- Implemented streaming pipelines with **Apache Kafka** and **Amazon Kinesis** to ingest trade lifecycle events for timely compliance monitoring and operational analysis.
- Created modular dbt transformations and **Great Expectations** validation frameworks with unit and **automated integration testing** for governed financial datasets.
- Tuned **Apache Spark** workloads, **SQL Server** queries, indexing strategies, and database security controls to improve processing efficiency, stability, and reliability.
- Collaborated with architects, analytics teams**, and business leaders to document data flows, define quality requirements, and implement solutions aligned with enterprise standards.
- Maintained an Amazon S3-based **Delta Lake** using **medallion architecture** and automated infrastructure and pipeline deployments through Terraform and Jenkins.

Environment: SQL, AWS Glue, Amazon EMR, Amazon S3, Amazon Kinesis, Apache Spark, PySpark, Delta Lake, Apache Kafka, Snowflake, SQL Server, dbt, Star Schema, Slowly Changing Dimensions

Data Engineer

- Developed and maintained **Azure Data Factory** pipelines integrating claims, eligibility, and provider data for healthcare analytics and operational reporting.
- Built **PySpark** ETL pipelines on **Databricks**, integrating EHR systems and third-party healthcare datasets into analytics-ready formats supporting clinical and operational insights.
- Designed **Snowflake** data warehouse models supporting claims analytics, utilization reporting, and clinical and operational dashboards.
- Built **streaming pipelines using Event Hub** and Structured Streaming, enabling near real-time tracking of claims processing and operational events.
- Optimized **Spark workloads and cluster configurations**, reducing processing time by ~25–30% across large healthcare datasets.
- Designed and maintained an **ADLS Gen2** data lake using **medallion architecture** and implemented governance with Azure Purview, enabling scalable processing, metadata lineage, and HIPAA-aligned data management.
- Developed **Power BI** dashboards for operations and clinical teams to monitor utilization trends, cost metrics, and key performance indicators.

Environment: Azure Data Factory, ADLS Gen2, Azure Event Hubs, Apache Spark, PySpark, Databricks, Snowflake, Microsoft Purview, Power BI, Medallion Architecture, Microsoft Azure

Target Corporation | Bangalore, India

Jan 2020 – Dec 2021

Data Engineer

- Developed AWS Glue and **Apache Airflow** ETL pipelines that integrated retail sales and inventory data from POS and e-commerce systems into **Amazon S3**.
- Built **PySpark** pipelines combining product, pricing, and order data into unified datasets for merchandising and demand analytics.
- Designed **Amazon Redshift Star Schema** models supporting sales performance, inventory tracking, and store-level reporting.
- Implemented **Apache Kafka** ingestion pipelines for near-real-time POS data used to improve inventory visibility and operational insights.
- Created **Python** and **SQL** validation checks to improve the reliability of transactional retail datasets.
- Optimized SQL queries and **indexing** in **PostgreSQL/MySQL**, improving performance of BI and reporting workloads.
- Developed **Tableau** dashboards enabling merchandising and operations teams to analyze product demand, customer behavior, and store performance.

Environment: Python, SQL, AWS Glue, Amazon S3, Amazon Redshift, PySpark, Apache Kafka, PostgreSQL, MySQL, Apache Airflow, Star Schema, Tableau, Indexing, AWS

EDUCATION

Master of Science in Computer Science

Rowan University | Glassboro, NJ

Jan 2024 – Dec 2025